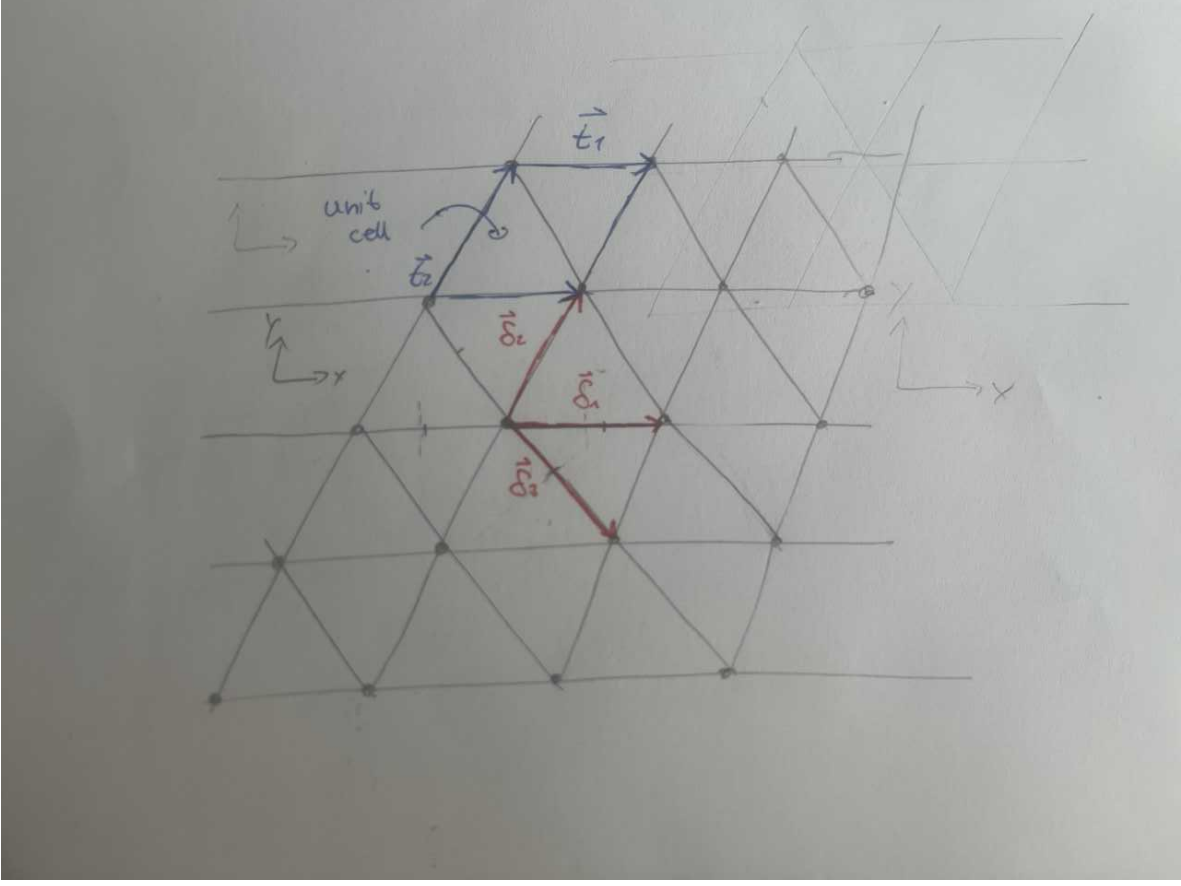


0.1 0-0 and π - π States



This system is a simple triangular Bravais lattice with primitive translations

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix}. \quad (1)$$

0.1.1 Pedestrian method

The tight-binding Hamiltonian is:

$$\begin{aligned} H &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f_{i\sigma}^\dagger f_{i+\delta\sigma} + \text{h.c.} \right) + \mu \sum_i \left(f_{i\uparrow}^\dagger f_{i\uparrow} + f_{i\downarrow}^\dagger f_{i\downarrow} - 1 \right) + \text{const.} \\ &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f_{i\sigma}^\dagger f_{i+\delta\sigma} + \text{h.c.} \right) + \mu \sum_{i,\sigma} f_{i\sigma}^\dagger f_{i\sigma} - \mu N + \text{const..} \end{aligned} \quad (2)$$

i runs over sites and δ over nearest neighbours of a generic site. I include the absolute shift for the sake of consistency. The factor 1/2 is also included for consistency. The constant is a product of mean field theory and reads

$$\frac{1}{2} \sum_{i,\delta} 2\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 = 6N \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) \right) |\langle \chi_{i,i+\delta} \rangle|^2 \quad (3)$$

The second step can be understood in at least two ways; a) each site has 6 neighbours, or b) there are bond per primitive unit cell 1 of which is completely inside the unit cell and contributes a 2 and the

other 4 shared between 2 unit cells thereby contributing a 1. All bonds are equivalent of course. The aforementioned six nearest neighbours lie at the (relative) positions

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \quad \vec{\delta}_3 = \begin{pmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

and $-\vec{\delta}_1, -\vec{\delta}_2, -\vec{\delta}_3$. I will start with the 0-0 state. To advance, do a Fourier transform:

$$f_{i\sigma} = \frac{1}{\sqrt{N}} \sum_k e^{i\vec{k}\vec{R}_i} c_{k\sigma}. \quad (4)$$

$$\begin{aligned} \Rightarrow H &= \sum_{q,\sigma} \frac{t}{2} \left(c_{q\sigma}^\dagger c_{q\sigma} \left(\sum_{\delta} e^{i\vec{q}\vec{\delta}} + \mu \right) + \text{h.c.} \right) - \mu N \\ &= \sum_{q,\sigma} \left(t \sum_{\delta} \cos(\vec{q}\vec{\delta}) + \mu \right) c_{q\sigma}^\dagger c_{q\sigma} - \mu N \end{aligned} \quad (5)$$

Using first the symmetrie of the cosine and then the sum identity for the cosine the dispersion reads

$$\begin{aligned} \varepsilon(\vec{q}) &= 2t \left[\cos(q_x) + \cos\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) + \cos\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) \right] + \mu \\ &= 2t \left[\cos(q_x) + 2 \cos\left(\frac{q_x}{2}\right) \cos\left(\frac{\sqrt{3}q_y}{2}\right) \right] + \mu \end{aligned} \quad (6)$$

features:

- $\varepsilon(\vec{q}) \in [-3t, 6t] + \mu \Rightarrow \mu \in [-6t, 3t]$
- symmetries $q_x \rightarrow -q_x$ and $q_y \rightarrow -q_y$ (+ should have some rotational symmetrie and (thereby associated) inflections along other axes) **could check this**
- periodicity: 4π in q_x , $(4\pi)/\sqrt{3}$ in q_y

The mean field parameters are

$$\langle \chi_{ij} \rangle = \frac{1}{2} \langle f_{i\uparrow}^\dagger f_{j\uparrow} + f_{i\downarrow}^\dagger f_{j\downarrow} \rangle = \langle f_{i\uparrow}^\dagger f_{j\uparrow} \rangle \quad (7)$$

Simplify the expression by Fourier transform:

$$\begin{aligned} \langle \chi_{ij} \rangle &= \frac{1}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{k}\vec{R}_i)} \langle c_p^\dagger c_q \rangle \\ &= \frac{1}{N} \sum e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q})) \end{aligned} \quad (8)$$

Take the thermodynamic limit $\sum_k \rightarrow (N\Omega)/(2\pi)^2 \int dk$ (Ω is the unit cell volume).

$$\Rightarrow \langle \chi_{ij} \rangle = \Omega \int_{\text{1.BZ}} \frac{d^2 q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q})) \quad (9)$$

It looks like the model allows for different χ_{ij} in each direction $\vec{R}_j - \vec{R}_i$ (there are 3 of those)?

From this expression, recover the on-site particle number by setting $i = j$ and accounting for the factor 1/2 in the definition of χ_{ij} :

$$\langle N_i \rangle = 2\Omega \int \frac{d^2 q}{(2\pi)^2} f(\varepsilon(\vec{q})). \quad (10)$$

By construction the particle number is uniform.

Some elaborations on the integral domain follow. The primitive translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = 2\pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad (11)$$

I do a basis transformation onto those axes defined by

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} \quad (12)$$

$$\Leftrightarrow \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \quad (13)$$

$$\Rightarrow \det J = -2/\sqrt{3} \quad (14)$$

$$\Rightarrow \varepsilon(\vec{k}) = 2t (\cos(k_x) + \cos(k_y) + \cos(k_x - k_y)) + \mu \quad (15)$$

The first Brillouin zone is $\{k_x, k_y\} \in [-\pi, \pi]^2$. The unit cell volume is $\Omega = \sqrt{3}/2$. Consistency check: $|\det(J)| = \Omega^{-1}$, since the substitution simply transforms the unit cell to a square of unit volume. So the integrals that have to be evaluated are of the form

$$\langle N_i \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} f(\varepsilon(\vec{k}), \mu), \quad (16)$$

$$\langle \chi_{ij} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, (1/\sqrt{3})(2k_x - k_y) \right) \cdot (\vec{R}_j - \vec{R}_i) \right] f(\varepsilon(\vec{k}), \mu) \quad (17)$$

Check that under a particle hole transformation χ changes sign, sign of t is doing just that and this brings the π -flux phase.

The condition is $\langle N_i \rangle \stackrel{!}{=} 1 \forall i$. By setting $t = -1$, the chemical potential is numerically determined in units of t using Mathematica. I choose $t = -1$ because I anticipate t to be negative for a positive χ . (This may not be true depending on κ and χ , but as long as it is self-consistent everything is fine.) For the numerical integration I used a Fermi-function with a small temperature ($\beta = 200t$) instead of a step-function, since the ladder is not analytical and causes problems for the numerical algorithm. The result is $\mu = -0.834737 t$. The bond-parameters are calculated the same way, the result is $\langle \chi_{ij} \rangle = 0.164713$.

Now, it is possible to express μ only in units of J . Use the formula

$$t_{ij} = -J \langle \chi_{ij} \rangle^* (1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2). \quad (18)$$

Set $\kappa = 10$:

$$t_{ij} = -0.8095 J \quad (19)$$

$$\mu = -0.67574 J \quad (20)$$

$$\langle \chi_{ij} \rangle = 0.164713 \quad (21)$$

The energy density (that is also the free energy density at $T = 0$) is

$$\frac{\langle H \rangle}{N} = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon(\vec{k}) f(\varepsilon(\vec{k})) - \mu + 6\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i, i+\delta} \rangle|^2 \right) |\langle \chi_{i, i+\delta} \rangle|^2 = -0.888304 J \quad (22)$$

Observation; the chemical potential drops out of the Free energy.

One can also directly calculate results for the state with all fluxes π by performing a particle-hole-transformation:

$$t \rightarrow -t \quad (23)$$

$$\mu \rightarrow -\mu \quad (24)$$

$$\Rightarrow \varepsilon(\vec{k}) \rightarrow -\varepsilon(\vec{k}) \quad (25)$$

$$\Rightarrow f(\varepsilon(\vec{k})) \rightarrow 1 - f(\varepsilon(\vec{k})) \quad (26)$$

It can be shown that

$$\chi \rightarrow -\chi. \quad (27)$$

The result is:

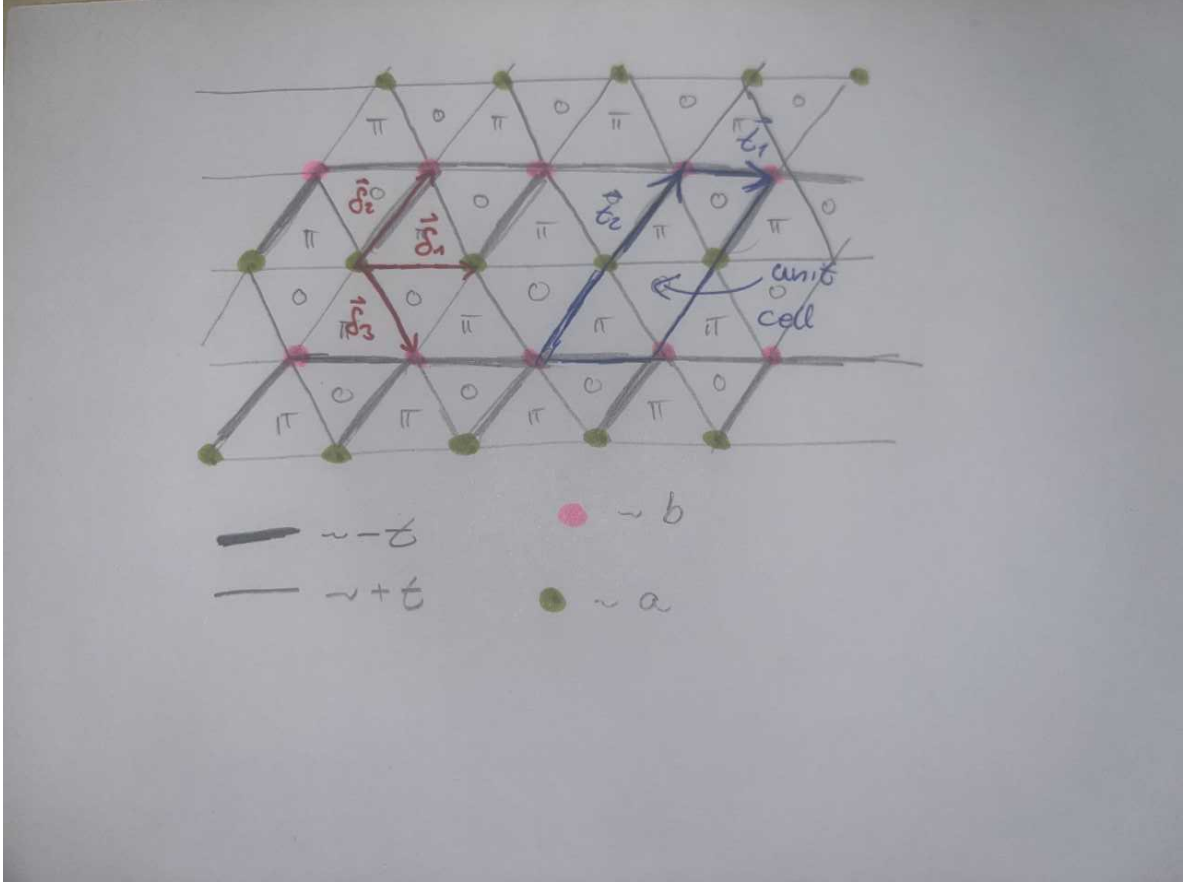
$$\frac{\langle H \rangle_\pi}{N} = -0.888302 J \quad (28)$$

0.1.2 Pedro's method

The longer explanation for this method is in the next section. Here I will only say that the free energy is the same except for γ , which gets replaced by a factor ϕ (as mentioned above the chemical potential does not contribute to the free energy):

0.2 0- π and π -0 States

0.2.1 pedestrian method



The system is a triangular bravais lattice with a two-atomic basis. The primitve translations are

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad (28)$$

Hence, two types of lattice sites exist, a and b (see sketch). Also χ_{ij} , and thereby t_{ij} can change sign (see sketch). The Hamiltonian is

$$H = \frac{1}{2} \sum_{\langle i,j \rangle, \sigma} \left(t a_{i\sigma}^\dagger a_{j\sigma} - t b_{i\sigma}^\dagger b_{j\sigma} + t_{ij}^{(ab)} a_{i\sigma}^\dagger b_{j\sigma} + \text{h.c.} \right) + \mu_a \sum_{i,\sigma} \left(a_{i\sigma}^\dagger a_{i\sigma} - 1 \right) + \mu_b \sum_{i,\sigma} \left(b_{i\sigma}^\dagger b_{i\sigma} - 1 \right) + \text{const.} \quad (29)$$

*a and b sites are connectd by bonds of both flavours. Mind that the sum above generates 24 terms. The constant is the same as for the 0-flux case, provided that the absolute value of $\chi_{i,i+\delta}$ is uniform.

$$\frac{1}{2} \sum_{i,\delta} 2\tilde{J} \left(1 - 6|\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 = 6N\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 \quad (30)$$

This is not quite right yet. The Fourier transformation has to be modified bc there are only N/2 of a and b-sites respectively

$$a_{i\sigma} = \sqrt{\frac{2}{N}} \sum_k e^{i\vec{k}\vec{R}_i} a_{k\sigma}, \quad (31)$$

and the same for $b_{i\sigma}$. The Hamiltonian (without the constants) becomes

$$H = \sum_{q,\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 2t \cos(\vec{q}\vec{\delta}_1) + \mu_a & 2t \cos(\vec{q}\vec{\delta}_3) - 2it \sin(\vec{q}\vec{\delta}_2) \\ 2t \cos(\vec{q}\vec{\delta}_3) + 2it \sin(\vec{q}\vec{\delta}_2) & -2t \cos(\vec{q}\vec{\delta}_1) + \mu_b \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix}. \quad (32)$$

The primitive translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = 2\pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad (33)$$

The suitable transformation is

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} 1 & \sqrt{3} \\ 1 & 0 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} \quad (34)$$

$$\Leftrightarrow \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \quad (35)$$

$$\Rightarrow \det(J) = 1/\sqrt{3} \quad (36)$$

This yields

$$\begin{aligned} \cos(\vec{q}\vec{\delta}_1) &\rightarrow \cos(k_y) \\ \sin(\vec{q}\vec{\delta}_2) &\rightarrow \sin(k_x/2) \end{aligned} \quad (37)$$

$$\cos(\vec{q}\vec{\delta}_3) \rightarrow \cos(k_y - k_x/2) \quad (38)$$

The dispersion relation is

$$\varepsilon(\vec{k})_{\pm} = \mu_a + \mu_b \pm \sqrt{(\mu_a - \mu_b)^2 + 2t \cos(k_y)(\mu_a - \mu_b) + 4t^2 \cos^2(k_y) + 4t^2 \cos^2(k_y - k_x/2) + 4t^2 \sin^2(k_x/2)} \quad (39)$$

Apparently, half filling is achieved globally, iff $\mu_a = -\mu_b \equiv \mu$. (The spectrum is symmetric around $\varepsilon = 0$ in this case.) In order to fulfill the local constraint one still has to calculate the local expectation values. It would be weird however, if the chemical potential differed between a and b sites (-i charge density waves?).

$$\Rightarrow \varepsilon(\vec{k})_{\pm} = \pm 2|t| \sqrt{(\mu/(2t) + \cos(k_y))^2 + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \quad (40)$$

It is correct that ε only depends on the absolute value of t , it looks like this would completely neutralise the particle-hole transformation. The periodicity is a bit weird here, since the expression is not periodic in 2π in every direction as one would expect. I am pretty sure this comes from the fact that the distance between NN is the same as for the 0-flux case but the Brillouin zone got cut in half in one direction. represent hamiltonian as

$$H = \sum_{q,\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger \end{pmatrix} \vec{h} \cdot \vec{\sigma} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix} \quad (41)$$

$$\vec{h} = \begin{pmatrix} 2t \cos(k_y - k_x/2) \\ 2t \sin(k_x/2) \\ 2t \cos(k_y) + \mu \end{pmatrix} \quad (42)$$

Then $\varepsilon(\vec{q})_{\pm} = \pm |\vec{h}|$ and

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t \cos(k_y) + \mu)}} \begin{pmatrix} \varepsilon_+ + 2t \cos(k_y) + \mu & -2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) \\ 2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) & \varepsilon_+ + 2t \cos(k_y) + \mu \end{pmatrix} \quad (43)$$

is the transformation matrix U . It has the properties that it is unitary and has determinant one. The expectation values are more involved bc of said matrix transformation. For example

$$\begin{aligned} \langle a_{i\uparrow}^\dagger b_{j\uparrow} \rangle &\stackrel{\text{FT}}{=} \frac{2}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{p}\vec{R}_i)} \underbrace{\langle a_p^\dagger b_q \rangle}_{\propto \delta_{q,p}} \\ &= \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \langle a_q^\dagger b_q \rangle \\ &\stackrel{\text{BT}}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} (u_{11}^* u_{21} f(\varepsilon_+) + u_{12}^* u_{22} f(\varepsilon_-)) \end{aligned} \quad (44)$$

Simplify this using

1. $\varepsilon_+ = -\varepsilon_- \Rightarrow f(\varepsilon_+) + f(\varepsilon_-) = 1$
2. $u_{11} \in \mathbb{R}$
3. $u_{11} = u_{22}$
4. $u_{12}^* = -u_{21}$
5. $|u_{11}|^2 + |u_{12}|^2 = 1$ (normalization of eigenvectors)

$$\begin{aligned} u_{11}^* u_{21} f(\varepsilon_+) + u_{12}^* u_{22} f(\varepsilon_-) &\stackrel{1.}{=} (u_{11}^* u_{21} - u_{22} u_{12}^*) f(\varepsilon_+) + u_{22} u_{12}^* \\ &\stackrel{2.,3.,4.}{=} 2u_{11} u_{21} f(\varepsilon_+) - u_{11} u_{21} \\ &= u_{11} u_{21} (2f(\varepsilon_+) - 1) \end{aligned} \quad (45)$$

Due to the unit cell doubling the thermodynamic limit has to be slightly modified $\sum_k \rightarrow ((\Omega N)/2)/(2\pi)^2 \int dk$. (Each unit cell has 2 sites, this is why the factor 1/2 appears.)

$$\begin{aligned} \Rightarrow \langle a_{i\uparrow}^\dagger b_{j\uparrow} \rangle &= \Omega \int_{\text{1.BZ}} \frac{d^2 q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} u_{11} u_{21} (2f(\varepsilon_+) - 1) \\ &= \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] u_{11} u_{21} (2f(\varepsilon_+) - 1) \end{aligned} \quad (46)$$

$$u_{11} u_{21} = (t/\varepsilon)(\cos(k_y - k_x/2) + i \sin(k_x/2)) \quad (47)$$

Now for the other expectation values:

$$\Rightarrow \langle b_{i\uparrow}^\dagger a_{j\uparrow} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] u_{11} u_{21}^* (2f(\varepsilon_+) - 1) \quad (48)$$

$$\begin{aligned}
\langle a_{i\uparrow}^\dagger a_{j\uparrow} \rangle &= \dots = \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [|u_{11}|^2 f(\varepsilon_+) + |u_{12}|^2 f(\varepsilon_-)] \\
&\stackrel{1.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [|u_{12}|^2 - |u_{11}|^2] f(\varepsilon_-) + |u_{11}|^2 \\
&\stackrel{5.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [(1 - 2u_{11}^2) f(\varepsilon_-) + u_{11}^2] \\
&\stackrel{1.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \tag{49}
\end{aligned}$$

$$\Rightarrow \langle a_{i\uparrow}^\dagger a_{j\uparrow} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \tag{50}$$

$$\Rightarrow \langle b_{i\uparrow}^\dagger b_{j\uparrow} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] [(1 - 2u_{11}^2) f(\varepsilon_+) + u_{11}^2] \tag{51}$$

$$u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t \cos(k_y) + \mu) \tag{52}$$

$$\langle N^a \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} (1 - 2u_{11}^2) f(\varepsilon_-) + u_{11}^2 \tag{53}$$

$$\langle N^b \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} (1 - 2u_{11}^2) f(\varepsilon_+) + u_{11}^2 \tag{54}$$

The constraint $\langle N_i^a \rangle = 1 = \langle N_i^b \rangle$ **should**, as gloal halffilling is already ensured and $\langle N^a \rangle$ and $\langle N^b \rangle$ are uniform by construction, be equivalent to $\langle N^a \rangle = \langle N^b \rangle$. Using the expressions above:

$$\langle N^a \rangle - \langle N^b \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} (1 - 2u_{11}^2)(1 - 2f(\varepsilon_+)) \stackrel{!}{=} 0 \tag{55}$$

Numerics say that $\langle N^a \rangle = \langle N^b \rangle$ for $\mu = 0$ (and therefore $\mu_a = \mu_b = 0$). **I feel like there is a more suphisticated argument to be made here, mabye even without the use of mathematica. Maybe one can find some symmetry argument using the form of $1 - 2f(\varepsilon_+)$.**

Sanity check:

$$\langle N^a + N^b \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} 1 = 2 \tag{56}$$

Calculating the mean field parameters is less straight forward, as the formulae also depend on the types of sites they connect. The formulae are

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \quad (57)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)f(\varepsilon_+) + u_{11}^2] \quad (58)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_3\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{i(k_y - k_x/2)} u_{11} u_{21} (2f(\varepsilon_+) - 1) \quad (59)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_2\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_x/2} u_{11} u_{21}^* (2f(\varepsilon_+) - 1) \quad (60)$$

Consistency check: wehn reversing the direction of the bonds the $\vec{\delta}_2$ bond paramter should obtain a minus sign but the $\vec{\delta}_3$ bond parameter not. **I wasnt able to figure this out by just looking at the formulae, so Ill just have to check by looking at the numerics for now.**

Another observation:

$$\begin{aligned} \frac{1}{2} \sum_{\sigma} \left(\langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle + \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle \right) &= \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_y} \\ &= \frac{1}{2\pi} \int_{[-\pi, \pi]} dk_y e^{ik_y} \\ &= \frac{\sin(\pi)}{\pi} \\ &= 0 \end{aligned} \quad (61)$$

$$\Rightarrow \frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = -\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle, \quad (62)$$

as expected. The numerical results are

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = 0.19754 \quad (63)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle = -0.19754 \quad (64)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_3\sigma} \rangle = 0.19754 \quad (65)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_2\sigma} \rangle = 0.19754 \quad (66)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i-\delta_2\sigma} \rangle = -0.19754 \quad (67)$$

The corresponding hopping amplitudes are

$$t(\pm 0.19754) = \mp 0.87690J \quad (68)$$

$$(69)$$

At last I will derive the energy density (since I had trouble with this before).

$$\begin{aligned}
\frac{\langle H \rangle}{N} &= \frac{1}{N} \left\langle \sum_{q\sigma} \varepsilon_{+\sigma} \alpha_{+\sigma}^\dagger \alpha_{+\sigma} + \varepsilon_{-\sigma} \alpha_{-\sigma}^\dagger \alpha_{-\sigma} \right\rangle \\
&\xrightarrow{\text{th. lim.}} \frac{1}{N} \frac{\Omega N}{2} \int_{\text{1.BZ}} \frac{d^2 q}{(2\pi)^2} \varepsilon_+ f(\varepsilon_+) + \varepsilon_- f(\varepsilon_-) \\
&= \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon_+ (2f(\varepsilon_+) - 1)
\end{aligned} \tag{70}$$

(the blue 2 in the second line is the spin degeneracy)

Observation: the energy density is invariant under particle-hole transformation

$$\varepsilon_+ (2f(\varepsilon_+) - 1) \xrightarrow{\text{ph-trafo}} \varepsilon_- (2f(\varepsilon_-) - 1) = -\varepsilon_+ (1 - 2f(\varepsilon_+)) = \varepsilon_+ (2f(\varepsilon_+) - 1) \tag{71}$$

The full energy density is:

$$\frac{\langle H \rangle}{N} = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon_+(\vec{k}) (2f(\varepsilon_+(\vec{k})) - 1) + 6 \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 \right) = -1.2206 J \tag{72}$$

0.2.2 Pedro's method

Here I apply Pedros method using the terms from my Hamiltonian. The free energy is

$$f_0 \equiv \frac{F_0(\kappa, \chi, T=0)}{N} = \frac{E_0}{N} + \frac{1}{N} \sum_{\vec{k}} \varepsilon(\vec{k}) \tag{73}$$

$$E_0 = 6N \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) |\langle \chi_{ij} \rangle|^2 \right) \tag{74}$$

$$\varepsilon(\vec{k}) = -2 \left| -\tilde{J} \left(1 - 4\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) \langle \chi_{ij} \rangle \right| \sqrt{\cos^2(k_1) + \cos^2(k_3) + \sin^2(k_2)} \tag{75}$$

From now on I use the shorthand $|\langle \chi_{ij} \rangle| \equiv \chi$.

$$\begin{aligned}
\Rightarrow f_0 &= \tilde{J} (6(1 - 6\tilde{\kappa}\chi^2)\chi^2 - 2\gamma |\chi(1 - 4\tilde{\kappa}\chi^2)|) \\
&= \tilde{J} (\mp 2\gamma\chi + 6\chi^2 \pm 8\gamma\tilde{\kappa}\chi^3 - 36\tilde{\kappa}\chi^4)
\end{aligned} \tag{76}$$

with

$$\gamma = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \sqrt{\cos^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \approx 1.18524 \tag{77}$$

Compared to Pedros results there is also a factor 2, besides other problems, because Pedro calculates $f/M (f/2)!$

$$\frac{\partial f_0}{\partial \chi} = \tilde{J} (\pm 2\gamma - 12\chi) (12\tilde{\kappa}\chi^2 - 1) \quad \frac{\partial^2 f_0}{\partial \chi^2} = \tilde{J} (12 \pm 48\gamma\tilde{\kappa}\chi - 432\tilde{\kappa}\chi^2) \tag{78}$$

The extremal points of the free energy are at

$$\chi' = \gamma/6, \quad \chi'' = \pm 1/\sqrt{12\tilde{\kappa}} \tag{79}$$

$-1/\sqrt{12\tilde{\kappa}}$ is always a Maximum. $\gamma/6$ is a minimum for $\kappa \leq 6/\gamma^2$, afterwards $1/\sqrt{12\tilde{\kappa}}$ becomes a minimum:

$$f_0 = \begin{cases} \frac{\tilde{J}\gamma^2}{6} \left(\frac{\tilde{\kappa}\gamma^2}{18} - 1 \right), & \tilde{\kappa} \leq 6/\gamma^2 \\ \tilde{J} \left(\frac{1}{4\tilde{\kappa}} - \frac{2\gamma}{9} \sqrt{\frac{3}{\tilde{\kappa}}} \right), & \tilde{\kappa} > 6/\gamma^2 \end{cases} \quad (80)$$

$6/\gamma^2 \approx 2.1355$. For $\kappa = 10 \Rightarrow \tilde{\kappa} = 1.6$ and therefore $\tilde{\kappa} < 6/\gamma^2$. Hence, the free energy minimum is at

$$\chi = \gamma/6 = 0.19754, \quad (81)$$

with the free energy

$$f_0 = -1.22206 J. \quad (82)$$

These correspond to the values I calculated using the pedestrian method above. Mind that under particle-hole transformation the calculation doesn't change (the reason basically being the absolute value in the energy, or more physically the particle-hole symmetry of the system).

0.2.3 low energy physics

The dispersion relation reads

$$\varepsilon(\vec{q}) = 2|t| \sqrt{\cos^2(q_x) + \sin^2 \left(\frac{1}{2} (q_x + \sqrt{3}q_y) \right) + \cos^2 \left(\frac{1}{2} (q_x - \sqrt{3}q_y) \right)} \quad (83)$$

I will start by calculating the fermi surface $\varepsilon(\vec{q}) = 0$, which means that the expression under the square root must vanish. Using

$$\sin^2 \left(\frac{1}{2} (q_x + \sqrt{3}q_y) \right) + \cos^2 \left(\frac{1}{2} (q_x - \sqrt{3}q_y) \right) = 1 + \sin(q_x) \sin(\sqrt{3}q_y) \quad (84)$$

I obtain the following equation for the fermi surface:

$$\underbrace{1 + \cos^2(q_0^x)}_{\in[1,2]} = - \underbrace{\sin(q_0^x) \sin(\sqrt{3}q_0^y)}_{\in[-1,1]}. \quad (85)$$

The equation can be simplified by comparing the ranges of both sides (see above)

$$\Rightarrow 1 + \cos^2(q_0^x) \stackrel{!}{=} 1 \Rightarrow q_0^x = \frac{\pi}{2} + \pi k, \quad k \in \mathbb{Z} \quad (86)$$

$$\Rightarrow \sin(\sqrt{3}q_0^y) = \begin{cases} -1 & k \text{ even} \\ 1 & k \text{ odd} \end{cases} \quad (87)$$

The solution (and thereby the Fermi surface) is

$$q_0^x = \frac{\pi}{2} + \pi k \quad (88)$$

$$q_0^y = \frac{1}{\sqrt{3}} \left(\frac{(-1)^{k+1}\pi}{2} + 2\pi m \right), \quad (89)$$

where $k, m \in \mathbb{Z}$. This means that the fermi surface is a collection of isolated points. More scientifically spoken: the state is a semi-metal. Two of those Fermi-points lie in the first Brillouin zone:

$$\vec{K} = \frac{\pi}{2} \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad (k = m = 0) \quad (90)$$

$$\vec{K}' = \frac{\pi}{2} \begin{pmatrix} -1 \\ 1/\sqrt{3} \end{pmatrix} \quad (k = -1, m = 0). \quad (91)$$

Mind you that these points are not equivalent! Evidently they are time-reversal partners. To access the low-energy physics we must approximate the Hamiltonian around the fermi-points using

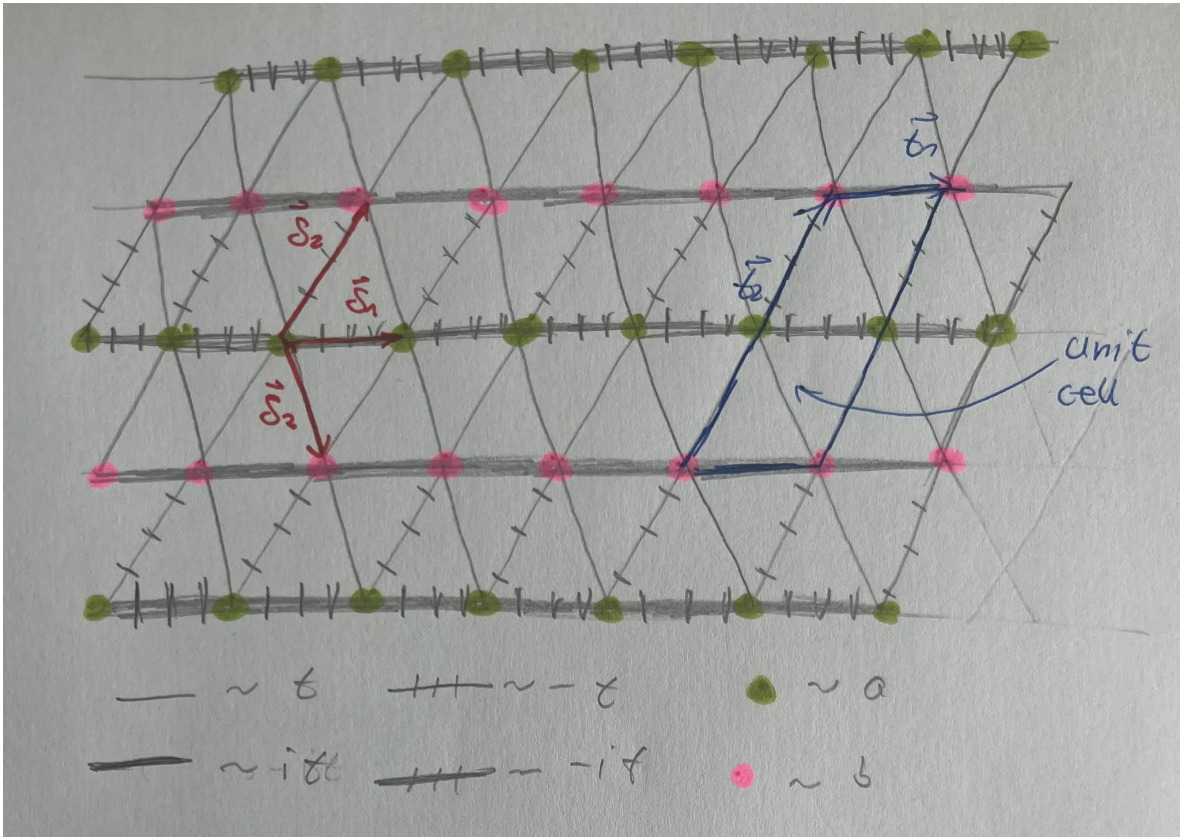
$$f(\vec{q}_0 + \delta\vec{q}) \approx f(\vec{q}_0) + (\nabla_{\vec{q}} f(\vec{q})|_{\vec{q}=\vec{q}_0}) \cdot \delta\vec{q} \quad (92)$$

$$\cos(q_x) \approx 0 + (-1)^k \delta q_x \quad (93)$$

$$\sin\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) \approx 0 + \frac{1}{2}\delta q_x + \frac{\sqrt{3}}{2}\delta q_y \quad (94)$$

$$\cos\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) \approx 0 + (-1)^k \left(-\frac{1}{2}\delta q_x + \frac{\sqrt{3}}{2}\delta q_y\right) \quad (95)$$

0.3 $\pi/2$ - $\pi/2$ and $-\pi/2$ - $-\pi/2$ States



0.3.1 Pedestrian Method

This is very similar to the $\pi - 0$ -flux-state. The bare Hamiltonian is

$$H = \sum_{i,\sigma} \left(\frac{t}{2} \left[-ia_{i\sigma}^\dagger a_{i+\delta_1;\sigma} + ia_{i\sigma}^\dagger a_{i-\delta_1;\sigma} - a_{i\sigma}^\dagger b_{i+\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i+\delta_3;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_3;\sigma} \right. \right. \\ \left. \left. + ib_{i\sigma}^\dagger b_{i+\delta_1;\sigma} - ib_{i\sigma}^\dagger b_{i-\delta_1;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_2;\sigma} - b_{i\sigma}^\dagger a_{i-\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i+\delta_3;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_3;\sigma} \right] \right. \\ \left. + \mu_a a_{i\sigma}^\dagger a_{i\sigma} + \mu_b b_{i\sigma}^\dagger b_{i\sigma} \right) - \frac{N(\mu_a + \mu_b)}{2} \quad (96)$$

$$\Rightarrow H = \sum_{q,\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 2t \sin(\vec{q}\vec{\delta}_1) + \mu_a & 2t \cos(\vec{q}\vec{\delta}_3) - 2it \sin(\vec{q}\vec{\delta}_2) \\ 2t \cos(\vec{q}\vec{\delta}_3) + 2it \sin(\vec{q}\vec{\delta}_2) & -2t \sin(\vec{q}\vec{\delta}_1) + \mu_b \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix}. \quad (97)$$

Maybe I should show explicitly that $\mu = 0$, I guess $\mu_a = -\mu_b = \mu$ is clear

Using $\mu_a = \mu_b = 0$:

$$\vec{h} = 2t \begin{pmatrix} \cos(\vec{q}\vec{\delta}_1) \\ \sin(\vec{q}\vec{\delta}_2) \\ \sin(\vec{q}\vec{\delta}_3) \end{pmatrix} \mapsto 2t \begin{pmatrix} \cos(k_y - k_x/2) \\ \sin(k_x/2) \\ \sin(k_y) \end{pmatrix} \quad (98)$$

$$\varepsilon(\vec{k})_\pm = \pm 2|t| \sqrt{\sin^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \quad (99)$$

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t \sin(k_y))}} \begin{pmatrix} \varepsilon_+ + 2t \sin(k_y) & -2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) \\ 2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) & \varepsilon_+ + 2t \sin(k_y) \end{pmatrix} \quad (100)$$

$$u_{11}u_{21} = (t/\varepsilon_+)(\cos(k_y - k_x/2) + i \sin(k_x/2)) \quad (101)$$

$$u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t \sin(k_y)) \quad (102)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^\dagger a_{i+\delta_1\sigma} \rangle = \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \quad (103)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^\dagger b_{i+\delta_1\sigma} \rangle = \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)f(\varepsilon_+) + u_{11}^2] \quad (104)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^\dagger b_{i+\delta_3\sigma} \rangle = \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{i(k_y - k_x/2)} u_{11}u_{21}(2f(\varepsilon_+) - 1) \quad (105)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^\dagger b_{i+\delta_2\sigma} \rangle = \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_x/2} u_{11}u_{21}^*(2f(\varepsilon_+) - 1) \quad (106)$$

One has to be a bit careful of where the sign of t enters when choosing $t = 1$ for the self consistency. Due to the complex conjugation the σ_z -terms are insensitive to the sign of t , but the (real) σ_x - and σ_y -terms are not. The results (choosing $t = 1$ are):

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = -0.200169i \quad (107)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i-\delta_1\sigma} \rangle = 0.200169i \quad (108)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle = 0.200169i \quad (109)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i-\delta_1\sigma} \rangle = -0.200169i \quad (110)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_3\sigma} \rangle = 0.200169 \quad (111)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i-\delta_3\sigma} \rangle = 0.200169 \quad (112)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_2\sigma} \rangle = 0.200169 \quad (113)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i-\delta_2\sigma} \rangle = -0.200169 \quad (114)$$

This complies with the cartoon on top. The value for t is

$$|t_{ij}| = \left| J \langle \chi_{ij} \rangle^* (1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2) \right|^{\kappa=10} \equiv 0.88020 J. \quad (115)$$

The free energy is

$$\frac{\langle H \rangle}{N} = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon_+(\vec{k}) (2f(\varepsilon_+(\vec{k})) - 1) + 6 \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 \right) = -1.24978 J \quad (116)$$

0.3.2 Pedro's Method

The expression for the free energy is identical to the $0 - \pi$ -state except that γ gets replaced by a different constant α :

$$\alpha = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \sqrt{\sin^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \approx 1.20102 \quad (117)$$

Crucially α is larger than gamma. The results for this method are

$$|\chi| = \alpha/6 = 0.20017, \quad (118)$$

$$f_0 = \tilde{J} \left(\frac{1}{4\tilde{\kappa}} - \frac{2\alpha}{9} \sqrt{\frac{3}{\tilde{\kappa}}} \right) = -1.24979 J \quad (119)$$

The methods agree.

0.3.3 Chern number

Here I will calculate the Chern number associated with the state. The relevant formula for a 2-band system is

$$C = \frac{1}{4\pi} \int_{\text{1.BZ}} d^2q \hat{h} \cdot \left(\partial_{q_x} \hat{h} \times \partial_{q_y} \hat{h} \right), \quad (120)$$

where $\hat{h}$ is the normalized Bloch vector ([is this the correct terminology?](#)).

$$\vec{h} = 2t \begin{pmatrix} \cos\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) \\ \sin\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) \\ \sin(q_x) \end{pmatrix} \equiv 2t \begin{pmatrix} \cos(\delta_3) \\ \sin(\delta_2) \\ \sin(\delta_1) \end{pmatrix}, \quad |\vec{h}| \equiv \varepsilon(\vec{q}) \quad (121)$$

I first simplify [120](#) a bit using

$$\partial_\alpha \hat{h} = \frac{1}{|\vec{h}|} \left(\partial_\alpha \vec{h} - (\hat{h} \cdot \partial_\alpha \vec{h}) \hat{h} \right). \quad (122)$$

When applying this to [120](#), 3 out of the four terms that appear are proportional to $\hat{h} \times \hat{h} = \vec{0}$.

$$\Rightarrow C = \frac{1}{4\pi} \int_{\text{1.BZ}} d^2q [\varepsilon(\vec{q})]^{-3} \vec{h} \cdot \left(\partial_{q_x} \vec{h} \times \partial_{q_y} \vec{h} \right) \quad (123)$$

$$\partial_{q_x} \vec{h} = t \begin{pmatrix} -\sin(\delta_3) \\ \cos(\delta_2) \\ 2\cos(\delta_1) \end{pmatrix}, \quad \partial_{q_y} \vec{h} = \sqrt{3}t \begin{pmatrix} \sin(\delta_1) \\ \cos(\delta_2) \\ 0 \end{pmatrix} \quad (124)$$

$$\begin{aligned} \Rightarrow \vec{h} \cdot \left(\partial_{q_x} \vec{h} \times \partial_{q_y} \vec{h} \right) &= 4\sqrt{3}t^3 (-\cos(\delta_1) \cos(\delta_2) \cos(\delta_3) + \cos(\delta_1) \sin(\delta_2) \sin(\delta_3) - \sin(\delta_1) \cos(\delta_2) \sin(\delta_3)) \\ &= 4\sqrt{3}t^3 (\sin(\delta_1) \sin(\delta_2) \cos(\delta_3) - 1) \end{aligned} \quad (125)$$

$$\Rightarrow C = \frac{\sqrt{3}}{8\pi} \int_{\text{1.BZ}} d^2q \frac{\sin(\delta_1) \sin(\delta_2) \cos(\delta_3) - 1}{(\sin^2(\delta_1) + \sin^2(\delta_2) + \cos^2(\delta_3))^{3/2}} \quad (126)$$

For evaluation I use the transformation:

$$\begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \quad (127)$$

$$\Rightarrow C = \frac{1}{8\pi} \int_{[-\pi, \pi]^2} d^2k \frac{\sin(k_y) \sin(k_y - k_x/2) \cos(k_x/2) - 1}{(\sin^2(k_y) + \sin^2(k_y - k_x/2) + \cos^2(k_x/2))^{3/2}} \quad (128)$$

The result is $C = -1$.